

Weatherly — What Can and Cannot Be Implemented / Integrated

Feature / Integration	Possible?	Stage	Why / constraint
Citizen reporting via web form + WhatsApp	✓	In MVP	Core feature in the project documentation.
Auto event classification (flood, heatwave, storm, fog, dust, wind)	✓	In MVP	NLP models (IndicBERT / XLM-R) and Whisper for speech.
Cross-check against IMD radar / AWS / satellite	✓	In MVP	Depends on access to IMD data (APIs / products).
Four-status trust output + human review queue	✓	In MVP	Deterministic ML + rule pipeline, chosen for explainability.
Public (masked) and authority dashboards	✓	In MVP	Role-based access (citizen / analyst / authority / researcher).
Native mobile app with offline report queue	✓	Future	Standard approach: queue locally, sync when connected. Out of scope for this version.
More Indian languages (beyond Hindi / English)	✓	Future	Needs training/evaluation data per language; multilingual models exist.
CAP-compatible alert ingestion / publishing	✓	Future	CAP is an open standard used by SACHET, SAMBRO and NOAA.
MOSDAC / Bhuvan satellite and GIS layers	✓	Future	Public ISRO portals; check data-access terms before relying on them.
Deepfake / AI-media detection	✓	Future	Via a third-party detector API (e.g. Hive, Reality Defender); likely licensed/paid. Output is evidence, not proof.
Kafka + Spark + Elasticsearch scale architecture	✓	Future	Documented but not built; heavier infrastructure and ops effort.
Reporter reputation score	✓	Future	Must be transparent and privacy-safe.
Open API / anonymised research exports	✓	Future	Exportable historical data is already a stated researcher need.
Short-term hotspot prediction	✓	Future	Possible once verified-event history is large; must be a separate, clearly labelled module.
Live Twitter/X and Instagram ingestion	✗	Not feasible now	Paid/restricted APIs; MVP uses sample data instead.
Replacing IMD weather forecasting	✗	Out of scope	Weatherly verifies reports; official forecasting stays with IMD.
Issuing official public emergency broadcasts	✗	Out of scope	Only authorised agencies (e.g. NDMA / SACHET) can do this; Weatherly supports them with evidence.
Auto-firing high-impact alerts from a single report	✗	Blocked by design	Requires a corroborated cluster plus human approval.
Declaring a report 'fake' with certainty	✗	Blocked by design	Only 'flagged, manipulation suspected' with evidence; the system abstains when unsure.
Publicly naming or accusing a citizen of posting fake content	✗	Blocked by design	Forbidden by the guardrails and a privacy risk.
Hardware / IoT sensor deployment	✗	Out of scope (this version)	Could be revisited later as optional community sensors.
Legal / forensic evidence chain	✗	Out of scope	Weatherly output is decision support, not court-grade evidence.